

Executive Summary & Technical Report: Nutrition Consumption Insights Dashboard

Project Overview

This report documents the end-to-end process of designing and building the Nutrition Consumption Insights Dashboard using Power BI. The objective of this dashboard is to analyze demographic trends, dietary habits, calorie distribution, and hydration patterns across a sample population of 5,747 individuals.

1. Data Cleaning & Transformation (Power Query)

Before visual layout and analysis, the raw dataset underwent systematic preprocessing in Power Query Editor to ensure data integrity and query performance:

- **Handling Missing & Invalid Values:** Checked for nulls or negative entries across key numeric fields (e.g., Daily Calories, Water Intake). Standardized caloric boundaries.
- **Data Type Enforcement:** Explicitly set data types for key metrics:
- **Person_ID:** Text / Whole Number (Unique Identifier)
- **Age:** Whole Number
- **Daily Calories, Carbohydrate Intake, Water Intake:** Decimal / Numeric
- **Diet_Type, Health_Status, Activity_Level, Gender:** Text
- **Categorical Standardizations:** Standardized text fields to eliminate case sensitivity issues and inconsistent spacing (e.g., ensuring uniform casing across Balanced, High Protein, Vegetarian, Mediterranean, Keto, and Vegan).
- **Binning & Grouping:** Calculated age bins and structured categorical groupings (e.g., Obese, Overweight, Healthy, Underweight) to streamline cross-filtering.

2. Data Modeling & DAX Measures

To power the KPIs and dynamic visual interactions, custom DAX (Data Analysis Expressions) measures were created:

- Total Respondents Count
Total People = COUNTROWS ('Nutrition_Data')
- Maximum Calorie Intake
Maximum Calories = MAX ('Nutrition_Data'[Daily Calories])
- Minimum Calorie Intake
Minimum Calories = MIN ('Nutrition_Data'[Daily Calories])
- Average Daily Calories
Average Daily Calories = AVERAGE ('Nutrition_Data'[Daily Calories])

3. Core Visual Layout

- **KPI Scorecards:** Summarizes core metrics at a glance (5,747 Total People | 5,158 Max Calories | 1,000 Min Calories | 2.47K Avg. Daily Calories).
- **Diet Type by Persons (Horizontal Bar Chart):** Compares dietary distribution, highlighting Balanced (1,462) as the dominant preference, followed by High Protein (1,035) and Vegetarian (1,019).
- **Daily Calorie by Health Status (Column Chart):** Illustrates caloric density across weight categories (Obese ~2.3K, Overweight ~2.3K, Healthy ~2.2K, Underweight ~1.8K).
- **Carbohydrate Intake by Gender (Donut Chart):** Breaks down carb consumption ratio (57.28% Female vs. 42.72% Male).
- **Average Water Intake by Age (Line Chart):** Tracks fluid intake consistency across age cohorts (20 to 80+ years old).

4. Key Analytical Insights

- **Dietary Preferences:** Over 25% of the analyzed population follows a Balanced diet, while restrictive diets like Vegan (555) and Keto (690) represent the smallest segments.
- **Caloric Consumption vs. Health Status:** Calorie intake correlates directly with health classifications; individuals classified as obese or overweight consume an average of 2.3K calories/day, whereas Underweight individuals average 1.8K calories/day.
- **Gender Disparity in Carbohydrates:** Females account for a higher total volume of carbohydrate intake (941.63K / 57.28%) compared to males (702.16K / 42.72%).
- **Hydration Trends:** Average daily water intake remains relatively stable across age demographics, hovering between 2.66L and 2.91L per day, with minor variance across older cohorts.

5. Strategic Recommendations

- **Targeted Dietary Programs:** Given the high concentration of individuals in Obese and Overweight categories consuming ~2.3K daily calories, launch targeted calorie-reduction and portion-control nutrition guides focused on high-protein, balanced meal planning.
- **Underweight Caloric Intervention:** Implement weight-gain nutrition intervention strategies for the Underweight cohort to increase daily caloric density from 1.8K to healthier baseline thresholds.
- **Gender-Specific Macro Coaching:** Design gender-tailored macronutrient recommendations, focusing on balancing carbohydrate distribution among female demographic segments.